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Computer Science

LOWER SIXTH, COMPLETE LESSON NOTES

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Computer Science

Lower Sixth Complete Lesson Notes

Academic Calendar: 2026–2027 **Class:** Lower Sixth (All) **Based on the MINESEC Harmonised Progression Sheet, 122 Lessons**

These notes follow the official progression sheet chapter by chapter. Related "Part 1 / Part 2" lessons and Integration Activities have been combined into single topic sections so that each note is complete and ready to teach or revise from.

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TERM ONE

Chapter 1: Exploring AI Concepts and Emerging Technologies

Chapter Overview

This chapter introduces students to Artificial Intelligence (AI), one of the most important and rapidly developing fields in computer science today. Students will learn what AI is, how it is categorised, how Machine Learning works, how AI systems are developed, and why data is central to everything AI does. The chapter also examines the social, ethical, and practical implications of AI in society.

Learning Objectives

By the end of this chapter, students should be able to:

1. Define Artificial Intelligence and explain why it has become important in modern society.
2. Distinguish between the categories of AI by capability (Narrow, General, Super) and by function (Reactive, Limited Memory, Theory of Mind, Self-Aware).
3. Identify everyday examples of AI and emerging technologies linked to AI.
4. Define Machine Learning and distinguish it from traditional programming.
5. Explain the three main types of Machine Learning (supervised, unsupervised, reinforcement) with examples.
6. Describe key AI techniques including neural networks, deep learning, NLP, computer vision, and expert systems.
7. Outline the stages in developing an AI system.
8. Explain why data is important in AI, including issues of bias, privacy, and ethics.

9. Describe methods of data collection and the data analysis process.
10. Discuss the benefits and concerns/limitations of AI.

Key Terms

Term	**Definition**
Artificial Intelligence (AI)	Branch of computer science building systems that perform tasks normally requiring human intelligence
Narrow AI	AI designed for one specific task; all AI today is narrow AI
General AI (AGI)	Hypothetical AI with human-level intelligence across any task
Super AI	Hypothetical AI surpassing human intelligence in all respects
Reactive machine	AI that responds to current input only, no memory
Limited memory	AI that uses past data briefly to make decisions
Machine Learning (ML)	Subset of AI where systems learn patterns from data rather than being explicitly programmed

Term	Definition
Supervised learning	ML trained on labelled data (input paired with correct output)
Unsupervised learning	ML that finds patterns in unlabelled data on its own
Reinforcement learning	ML where an agent learns by trial and error, receiving rewards or penalties
Neural network	Layered structure loosely inspired by the brain, used for pattern recognition
Deep learning	Neural networks with many layers, powering image recognition and NLP
Natural Language Processing (NLP)	Enables computers to understand and generate human language
Computer Vision	Enables computers to interpret images and video
Expert System	Rule-based system that mimics a human expert's decision-making in a narrow domain
Training data	Data used to teach a machine learning model

Term	Definition
Test data	Data used to check a model's accuracy on unseen examples
Bias (in AI)	Systematic unfairness learned from biased training data
Big Data	Extremely large datasets that require special tools to analyse
IoT (Internet of Things)	Network of everyday objects with sensors connected to the internet
Primary data	Data collected directly for the purpose at hand
Secondary data	Data already collected by someone else
Data cleaning	Removing errors, duplicates, and missing values from data

1.1 Introduction to Artificial Intelligence

Definition: Artificial Intelligence (AI) is the branch of computer science concerned with building systems (hardware/software) that can perform tasks which normally require human intelligence, reasoning, learning, perception, problem-solving, and decision-making.

In simpler terms: AI is about making machines that can "think" and "learn" in ways that resemble human intelligence.

Why AI Matters Now

AI has existed as a concept since the 1950s, but it has only become practical in everyday life in the last two decades. Three factors made this possible:

1. Cheaper and more powerful computing power, modern processors and GPUs can perform billions of calculations per second at low cost.
2. Huge amounts of digital data ("Big Data"), billions of people generate data every day through phones, websites, sensors, and transactions. AI needs this data to learn.
3. Improved algorithms, especially deep learning, which has become far more effective since around 2012.

These three factors together have made AI practical in everyday devices: phones, cars, search engines, hospitals, farms, and homes.

Categories of AI (by Capability)

Category	**Description**	**Example**	**Exists today?**
Narrow / Weak AI	Designed for one specific task	Chess engine, spam filter, Siri, Google Translate	Yes, all AI in use today is narrow AI
General AI (AGI)	Hypothetical AI with human-level intelligence across any task	A machine that could learn and reason like a human across all domains	No, does not yet exist
Super AI	Hypothetical AI surpassing human	A machine far smarter than the best human in	No, theoretical only

Category	Description	Example	Exists today?
	intelligence in all respects	every field	

Important point: Despite what science fiction suggests, all AI today is narrow AI. Even the most advanced systems (like ChatGPT or self-driving cars) are designed for specific tasks and cannot generalize beyond them.

Categories of AI (by Function)

Category	Description	Example
Reactive machines	Respond to current input only; no memory of past events	IBM's Deep Blue (chess)
Limited memory	Use past data briefly to make decisions	Self-driving cars
Theory of mind	Would understand emotions, beliefs, and intentions of others (research stage)	—
Self-aware	Would have consciousness and a sense of self (theoretical)	—

Everyday Examples of AI

- **Voice assistants:** Siri, Alexa, Google Assistant
- **Recommendation systems:** Netflix, YouTube, TikTok, Amazon
- **Facial recognition:** phone unlock, security cameras, passport control
- **Spam filters:** email classification
- **Chatbots:** customer service on websites
- **GPS route optimization:** Google Maps, Waze
- **Medical diagnosis support:** X-ray and scan analysis
- **Agricultural drones:** crop monitoring, precision farming
- **Fraud detection:** banking and mobile money systems
- **Language translation:** Google Translate, DeepL

Emerging Technologies Linked to AI

Technology	**Description**
Internet of Things (IoT)	**Everyday objects with sensors connected to the internet**
Robotics	**Machines that perform physical tasks, often guided by AI**
Big Data analytics	**Extracting patterns from massive datasets**

Technology	Description
Cloud computing	**On-demand computing power and storage over the internet**
Autonomous vehicles	**Self-driving cars, drones, autonomous ships**
Blockchain	**Decentralised, tamper-resistant record-keeping**
Virtual/Augmented Reality (VR/AR)	**Immersive digital experiences**

How these technologies work together: IoT sensors on a farm collect soil and weather data → cloud computing stores it → AI analyses it → the farmer receives recommendations on when to water or harvest. This is often called the IoT–Cloud–AI pipeline.

Benefits of AI

- **Increased efficiency and speed**: AI can process data far faster than humans.
- **24/7 availability**: AI systems do not get tired or need breaks.
- **Reduced human error** in repetitive tasks.
- **Ability to process huge datasets**: beyond human capacity.
- **Improved accuracy** in prediction and diagnosis.
- **Cost savings** over time in automation.

Concerns / Limitations of AI

- **Job displacement**: automation of routine work (manufacturing, customer service, driving).
- **Bias in training data**: leading to unfair decisions (e.g., biased hiring or lending).
- **Lack of transparency**: the "black box" problem: we can't always explain why AI decided something.
- **Privacy risks**: AI often uses personal data.
- **High development cost**: building and maintaining AI systems is expensive.
- **Dependence on quality data**: poor data → poor AI.
- **Ethical questions**: autonomous decision-making (e.g., self-driving car accidents, autonomous weapons).
- **Security risks**: AI can be used for deepfakes, phishing, and cyberattacks.

1.2 AI Techniques and Machine Learning

What is Machine Learning?

Machine Learning (ML) is a subset of AI in which a system learns patterns from data rather than being explicitly programmed with fixed rules.

In traditional programming, a human writes the rules. In machine learning, the system discovers the rules itself by studying examples.

Traditional Programming vs Machine Learning

Traditional Programming	**Machine Learning**
Programmer writes explicit rules	System learns rules from data

Traditional Programming	Machine Learning
Input + Program → Output	Input + Output (training data) → Program (model)
Poor at handling unseen/complex patterns	Good at recognising complex patterns
Rules are transparent and editable	Rules are learned and often opaque
Works well for well-defined problems	Works well for pattern-recognition problems

Example: To filter spam traditionally, you'd write rules like "if the email contains 'free money', mark as spam." In ML, you'd show the system thousands of spam and non-spam emails, and it would learn the patterns itself.

Types of Machine Learning

1. Supervised Learning

The model is trained on labelled data, input paired with the correct output.

- **Classification**, predicting a category (spam / not spam; cat / dog; pass / fail).
- **Regression,** predicting a continuous value (house price, temperature, salary).

Example: A model trained on 10,000 emails labelled "spam" or "not spam" learns to classify new emails.

2. Unsupervised Learning

The model finds patterns in unlabelled data on its own**.**

- **Clustering:** grouping similar items (customers by buying behaviour).
- **Association:** finding rules (people who buy bread also buy butter).

Example: A supermarket's loyalty-card data is clustered into customer types without any pre-set labels.

3. Reinforcement Learning

An agent learns by trial and error, receiving rewards or penalties for actions.

Example: A robot learning to walk; a game-playing AI (AlphaGo); a self-driving car learning to stay in lane.

Comparison table:

Type	**Data**	**Goal**	**Example**
Supervised	Labelled	Predict output for new input	Spam filter, price prediction
Unsupervised	Unlabeled	Find hidden patterns	Customer segmentation
Reinforcement	Interaction + rewards	Learn optimal actions	Robot walking, game AI

Key AI Techniques

END OF FREE SAMPLE

Get the full notes

You have just read the opening of **Lower Sixth Computer Science**. The complete book has every lesson of the MINESEC progression sheet, in order:

- All 122 lessons, each with objectives, clear notes and KEY POINTS
- Worked examples and practice questions with answers
- Review exercises at the end of every chapter

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